Bihar Engineering University, Patna

B.Tech 6th Semester Examination, 2025

Course: B.Tech | ***Time: 03 Hours***
Code: 155608 | ***Subject: Natural Language Processing*** | ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
*(iv) Question No. **1** is compulsory.*

Q.1 Choose the correct option / answer the following (Any seven question only): [2 x

(a) In WordNet, the lexemes that share same form but have unrelated meanings is
(i) Homonym (iii) Hyponym
(ii) Hypernym (iv) Meronym

(b) Which of the text parsing techniques can be used for noun phrase detection, verb phrase detection, subject detection, and object detection in NLP.
(i) Part-of-Speech Tagging (iii) Continuous Bag-Of-Words
(ii) Skip Gram and n-gram Extraction (iv) Dependency Parsing and Constituency Parsing

(c) The study of the sound patterns in natural language and the rules that govern them is:
(i) Phonetics (iii) Phonology
(ii) Morphology (iv) Syntax

(d) The words "there" and "their" causes which of the following type of ambiguity?
(i) Syntactic (iii) Phonological
(ii) Semantic (iv) Pragmatic

(e) Which of the following is the instance of stemming as per the porter stemming algorithm?
(i) are-> be (iii) saw->s
(ii) Plays->play. (iv) university->universe

(f) Study of the meaning of words that are associated with grammatical structure is known as -
(i) Morphology (iii) Semantics
(ii) Phonology (iv) Reasoning

(g) Which is not the Classification levels in Sentiment Analysis?
(i) Document-level (iii) Aspect-level
(ii) Character-level (iv) Sentence-level

(h) Identify incorrect statement about Finite state Transducer (FST) from the following
(i) It is used as string recognizer (iii) It is used as translator
(ii) It is used as string generator (iv) It is to covert string into machine language

(i) Which of the following represents the statement "Which team won the match?"
(i) N->Wh-NP VP (iii) VP->Wh-NP VP
(ii) S->Wh-NP VP (iv) S->Wh-NP NP

(j) Which of the following transfers linear sequences of words into structure?
(i) Semantic Analysis (iii) Lexical analysis
(ii) Tokens (iv) Discourse

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Q.2	(a)	Define the term stemming. List the Porter Stemmer's steps 1a, 1b and 1c and stem some terms using these rule.	[7]
	(b)	Define and discuss the use of TF-IDF vectorizer with suitable example.	[7]
Q.3	(a)	Figure out whether *drive* is closer to *brief* or to *divers* and what the edit distance is to each. You may use Leveinstene edit distance technique.	[7]
	(b)	What is word boundary detection in the NLP? Discuss approaches used for detecting boundary briefly.	[7]
Q.4	(a)	Define Markov assumption in the language model. Explain, how it is derived from chain rule of probability?	[7]
	(b)	Why morphological analyses in Indian languages are more challenging? Explain it in detail	[7]
Q.5	(a)	What is Transition-Based dependency parsing? Draw the architecture of basic transition-based parser and explain its components.	[7]
	(b)	Explain the role of smoothing techniques in dealing with unseen words or phrases in N-Gram models	[7]
Q.6	(a)	What do you understand by word sense induction or WSI in word sense disambiguation? Discuss its application in detail.	[7]
	(b)	Discuss the role of HMM in speech recognition.	[7]
Q.7	(a)	Define the term Cross-Lingual Information Retrieval (CLIR). Why CLIR is important specially in India. Explain with example.	[7]
	(b)	Consider a corpus has three documents (d_1, d_2, and d_3). d_1: India is democratic country, d_2: Democratic country runs on principle of constitution d_3: Socialism term is amended in the preamble of our constitution (i) Create a term-document matrix of word 2-gram after handling required pre-processing steps from given data. (ii) From above matrix derive vectors (use count vector) to represent d_1, d_2, *and* d_3.	[7]
Q.8	(a)	Draw the general framework/architecture for document classification. Discuss different stages involved in it briefly.	[7]
	(b)	What Are Scope Ambiguities? Explain them briefly.	[7]
Q.9	(a)	Write short notes on (i) Sentiment analysis (ii) Scalable Machine translation	[7]
	(b)	Differentiate between top-down and bottom-up parsing in NLP	[7]

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